

APOLLO INSTITUTE OF COMPUTER APPLICATION

BACHELOR OF COMPUTER APPLICATION

SEMESTER 03

Design Thinking and Innovation (BC03001071)

Assignment 3

Unit 3: Defining Problems

QUESTION.1. Explain HMW questions in Know Your Problem. (Winter 2025)

ANSWER:

HMW stands for “**How Might We?**” It is a question format used in Design Thinking to convert a problem into an opportunity for finding creative solutions.

1. Meaning of HMW Questions

HMW questions are **open-ended questions** that help us look at a problem from a positive and solution-oriented perspective.

For example:

Problem: Students find programming difficult to understand.

HMW Question:

“How might we help students understand programming easily?”

2. Purpose of HMW Questions

HMW questions are used to:

1. Understand and clearly define the problem.
2. Focus on the **needs of the user**.
3. Encourage creative and innovative ideas.
4. Explore different possible solutions.
5. Connect **problem definition with ideation**.

3. How to Frame an HMW Question

An HMW question can be framed as:

How might we + user + problem/need + desired improvement?

Example:

Problem: Patients have to wait for a long time at a hospital.

HMW:

“How might we reduce the waiting time for patients in hospitals?”

4. Characteristics of a Good HMW Question

A good HMW question should be:

- **Open-ended:** Allows many possible answers.
- **User-focused:** Concentrates on the user's needs.
- **Clear:** Easy to understand.
- **Positive:** Focuses on opportunities for improvement.
- **Solution-neutral:** Does not directly suggest one solution.

5. Example

Problem: Students are not interested in attending lectures.

HMW Question:

“How might we make classroom lectures more interesting and engaging for students?”

Possible solutions could include interactive activities, games, discussions, demonstrations, etc.

Conclusion

HMW questions help designers **convert a problem into an opportunity**. They encourage creative thinking and help the team generate different solutions. Therefore, HMW questions are an important part of **defining problems in Design Thinking**.

QUESTION.2. Explain how to know your problem in detail. (Winter 2025)

ANSWER:

Know Your Problem is an important step in Design Thinking. It means **understanding the actual problem faced by the user before trying to find a solution**. A designer should not assume the problem; instead, the problem should be understood from the user's needs, experiences, and observations.

1. Identify the Problem

First, identify the problem that needs to be solved.

Example:

Students are finding it difficult to understand programming concepts.

2. Understand the User

Understand **who is facing the problem** and what difficulties they are experiencing.

For example, talk to students and find out:

- What difficulties do they face?
- Which topics are difficult?
- Why are they facing these difficulties?

3. Observe the Situation

Observe the users in their actual environment. Observation helps identify problems that users may not directly mention.

Example:

Observe students during programming practicals and notice where they face difficulties.

4. Collect Information

Collect information through:

- Interviews
- Surveys
- Observations
- User feedback

This helps the designer understand the problem from the user's point of view.

5. Find the Root Cause

Do not focus only on the visible problem. Find **why the problem is happening**.

Example:

Problem: Students cannot complete programming practicals.

Possible root cause: They may not understand basic programming concepts or may not get enough practice.

6. Define the Actual Problem

After collecting and analysing information, clearly define the **real user problem**.

For example:

“Students need a simple and practical way to understand programming concepts so that they can confidently solve programming problems.”

7. Convert the Problem into an HMW Question

Finally, the problem can be converted into a **How Might We (HMW)** question to encourage ideas.

Example:

“How might we help students understand programming concepts more easily through practical activities?”

Conclusion

Knowing your problem means understanding the user, observing the situation, collecting information, finding the root cause, and clearly defining the actual problem. A properly defined problem helps designers develop better and more user-centred solutions.

QUESTION.3. What is a Point-of-View (POV) statement in design thinking? Explain how it is constructed and describe its importance in the design process. *(Summer 2026)*

ANSWER:

A **Point-of-View (POV) statement** is a clear and focused statement that describes a **user, the user's need, and the key insight** gained from user research. It defines the problem from the **user's perspective** and helps designers develop suitable solutions.

1. Elements of a POV Statement

A POV statement consists of three main elements:

- **User:** Who is experiencing the problem?
- **Need:** What does the user need?
- **Insight:** Why does the user have this need?

Format:

[User] needs [Need] because [Insight].

2. How to Construct a POV Statement

The following steps are used:

1. **Identify the target user** from user research.
2. **Understand the user's needs** through interviews, observation, and feedback.
3. **Identify important insights** about the user's behaviour, difficulties, and expectations.
4. **Define the main problem** from the user's point of view.
5. **Combine the user, need, and insight** into one clear statement.
6. **Keep the statement specific and user-focused.**
7. **Avoid suggesting a solution** in the POV statement.

3. Example

Suppose research shows that college students find it difficult to remember assignment deadlines.

POV Statement:

A college student needs an easy way to track assignment deadlines because managing multiple subjects makes it difficult to remember important dates.

Here:

- **User:** College student
- **Need:** Easy way to track assignment deadlines
- **Insight:** Multiple subjects make it difficult to remember dates

4. Importance of POV Statement

1. **Defines the real problem:** It clearly identifies the problem faced by the user.
2. **Keeps the user at the center:** It focuses on the user's needs and experiences.
3. **Based on research:** It is developed using information collected during the empathy stage.
4. **Guides ideation:** It helps the team generate relevant ideas and solutions.
5. **Avoids assumptions:** It encourages designers to solve the actual user problem.
6. **Creates a common direction:** It helps all team members understand the same problem.
7. **Supports better solutions:** A clear POV helps create solutions that are useful and meaningful to users.

Conclusion

A **Point-of-View statement** is an important part of the **Define stage of Design Thinking**. It converts user research and insights into a **clear, user-centered problem statement** and provides direction for the **Ideation stage**.

QUESTION.4. Explain Insights in Design Thinking. How are insights identified and used to define a problem?

ANSWER:

Meaning of Insight

An **insight** in Design Thinking is a **meaningful understanding about the user's needs, behaviour, feelings, or problems** obtained through user research. It goes beyond simply collecting information and helps designers understand **why a user behaves in a particular way**.

For example, if students frequently miss assignment deadlines, the observation is “*students forget deadlines.*” The insight may be “*students find it difficult to manage deadlines when they have assignments from multiple subjects.*”

How are Insights Identified?

Insights are identified through the following steps:

1. **Conduct user research:** Collect information through interviews, observations, surveys, and discussions.
2. **Collect observations:** Record what users say, do, think, and feel.
3. **Find patterns:** Compare the collected information and identify repeated behaviours or problems.

4. **Identify user needs:** Understand what users actually need rather than only what they say they want.
5. **Find the “Why”:** Ask why the problem occurs to discover the underlying reason.
6. **Identify important findings:** Select findings that can influence the design of a solution.

How Insights are Used to Define a Problem

Insights help designers to:

1. **Understand the real problem** instead of focusing only on symptoms.
2. **Identify the user's actual needs and difficulties.**
3. **Remove assumptions** and base the problem on research.
4. **Create a clear Point-of-View (POV) statement.**
5. **Develop How Might We (HMW) questions** for generating ideas.
6. **Focus on the most important user problem.**
7. **Guide the ideation process** toward meaningful solutions.

Example

Observation: Students often forget assignment submission dates.

Insight: Students struggle to manage multiple deadlines because they do not have a simple system to organize their academic tasks.

Defined Problem:

College students need a simple way to organize and track assignment deadlines because managing multiple subjects makes it difficult to remember important dates.

Conclusion

Thus, **insights convert research findings into meaningful understanding of users**. They help designers discover the **root cause of a problem** and define a clear, user-centered problem that can be addressed through Design Thinking.

QUESTION.5. Explain the difference between facts, observations, and insights with suitable examples.

ANSWER:

In **Design Thinking**, facts, observations, and insights are different levels of understanding collected during the **Empathy and User Research** stage. They help designers move from **what is happening** to **why it is happening**.

1. Fact

A **fact** is objective information that can be **verified or measured**. It does not include personal opinions or assumptions.

Example:

“There are 50 students in a classroom.”

2. Observation

An **observation** is something that the designer **sees, hears, or notices** while studying the user's behaviour.

Example:

“Many students use their mobile phones during the lecture.”

3. Insight

An **insight** is a deeper understanding of **why the user behaves in a particular way**. It is derived by analysing facts and observations.

Example:

“Students use their mobile phones because they find it difficult to stay engaged when the lecture is too long and has limited interaction.”

Difference Between Fact, Observation and Insight

Basis	Fact	Observation	Insight
Meaning	Verifiable information	What the designer notices	Deeper understanding of the reason
Focus	What is true	What is happening	Why it is happening
Source	Data or records	Seeing/hearing users	Analysis of research
Nature	Objective	Descriptive	Interpretative
Example	50 students attend the class	Students frequently check phones	Students may lose engagement during lengthy lectures

Relationship

The three can be understood as:

Fact → Observation → Insight → Problem Definition

Example:

- **Fact:** 30% of students submit assignments late.
- **Observation:** Students often ask about submission dates.
- **Insight:** Students have difficulty tracking multiple deadlines.
- **Problem:** Students need an easy way to organize and remember assignment deadlines.

Conclusion

Facts tell us what is true, observations tell us what is happening, and insights explain why it may be happening. Insights are especially important in Design Thinking because they help designers identify the **real user problem** and create better solutions.

QUESTION.6. Explain HMW (How Might We) questions in detail. How are effective HMW questions framed?

ANSWER:

Questions Framed?

Meaning of HMW Questions

HMW stands for **How Might We**. HMW questions are open-ended questions used in the **Define and Ideation stages of Design Thinking**. They convert a **user problem or insight into an opportunity for generating ideas and solutions**.

Instead of directly asking “*What is the solution?*”, an HMW question asks “**How might we...?**”, which encourages designers to think of different possible solutions.

Example

Problem: Students find it difficult to remember assignment deadlines.

HMW Question:

How might we help students remember and manage their assignment deadlines easily?

This question does not provide one fixed solution. It allows designers to think of many ideas such as reminders, calendars, mobile applications, notifications, etc.

How Are Effective HMW Questions Framed?

An effective HMW question can be framed using the following steps:

1. Start with “How might we”

Begin the question with **How might we...** to create an open-ended and positive approach.

2. Focus on the user

Clearly identify the target user's need or difficulty.

3. Use insights from research

The question should be based on an actual user insight rather than assumptions.

4. Focus on the problem, not a specific solution

Do not include a particular solution in the question.

5. Keep it open-ended

The question should allow multiple possible answers and ideas.

6. Make it neither too broad nor too narrow

It should be specific enough to address a real problem but broad enough to encourage creativity.

7. Use positive language

Frame the problem as an opportunity rather than a limitation.

Example of Framing an HMW Question

User Insight:

Students find it difficult to manage multiple assignment deadlines.

Poor HMW Question:

How might we create a mobile app for assignment reminders?

This is too specific because it already suggests a solution.

Effective HMW Question:

How might we help students manage multiple assignment deadlines more easily?

This allows different solutions such as apps, calendars, reminder systems, notice boards, or other methods.

Characteristics of a Good HMW Question

A good HMW question should be:

- **User-centered**
- **Open-ended**
- **Clear and simple**
- **Based on research**
- **Solution-neutral**
- **Positive**
- **Focused on a specific problem**
- **Able to generate multiple ideas**

Importance of HMW Questions

1. They **turn problems into opportunities**.
2. They encourage **creative thinking**.
3. They generate **multiple possible solutions**.
4. They keep the focus on **user needs**.
5. They provide direction for the **ideation stage**.
6. They prevent the team from becoming attached to one solution.
7. They help the design team work together toward a common goal.

Conclusion

HMW questions are an important tool in Design Thinking for moving from problem definition to ideation. A well-framed HMW question is **user-centered, open-ended, positive, and solution-neutral**, allowing designers to explore many creative solutions to the identified problem.

QUESTION.7. Explain Point-of-View (POV) in Design Thinking. Explain its components, construction, and importance.

ANSWER:

Meaning of Point-of-View (POV)

A **Point-of-View (POV)** statement is a clear, user-centered statement that defines **who the user is, what the user needs, and why the need is important**. It is created after analysing user research and insights.

POV helps the design team focus on the **right problem from the user's perspective** before moving to the Ideation stage.

Components of a POV Statement

A POV statement mainly has **three components**:

1. User

- Identifies the specific person or group facing the problem.
- Example: *College students*

2. Need

- Describes what the user needs or wants.
- Example: *An easy way to manage assignment deadlines*

3. Insight

- Explains the deeper reason behind the user's need.
- Example: *Because students have difficulty managing deadlines for multiple subjects.*

Basic Format

[User] needs [Need] because [Insight].

Construction of a POV Statement

A POV statement can be constructed using these steps:

1. **Collect user information** through interviews, observations, surveys, etc.
2. **Identify user needs and problems** from the collected information.
3. **Analyse the research data** to find important patterns.
4. **Identify insights** that explain the underlying reasons behind the problems.

5. **Identify the target user** who is most affected by the problem.
6. **Combine the user, need, and insight** into a clear statement.
7. **Review the statement** to ensure that it is specific, user-centered, and does not directly suggest a solution.

Example

User Research: Students often miss assignment submission dates.

POV Statement:

A college student needs an easy way to track assignment deadlines because managing multiple subjects makes it difficult to remember important dates.

Here:

- **User:** College student
- **Need:** Easy way to track assignment deadlines
- **Insight:** Multiple subjects make it difficult to remember important dates.

Importance of POV

1. **Defines the real problem:** It clearly identifies the problem from the user's perspective.
2. **Keeps the user at the center:** It focuses on the user's needs and experiences.
3. **Uses research and insights:** It is based on actual information rather than assumptions.
4. **Guides ideation:** It provides direction for generating creative ideas.
5. **Avoids solution-based thinking:** It defines the problem without immediately choosing a solution.
6. **Creates a common understanding:** It helps the design team understand the problem clearly.
7. **Helps develop better solutions:** A well-defined POV increases the possibility of creating useful solutions.

Conclusion

A **POV statement connects user research with problem definition**. It combines the **user, need, and insight** into a clear statement and provides a strong foundation for the **Ideation stage** of Design Thinking.

QUESTION.8. Explain how to know your problem in detail. What are the important steps involved in defining the right problem?

ANSWER:

Meaning of “Know Your Problem”

In **Design Thinking**, “**Know Your Problem**” means developing a clear and deep understanding of the problem before trying to find a solution. Designers should identify **who is facing the problem, what the problem is, why it occurs, and how it affects the user**.

A designer should not immediately jump to a solution. First, the **real or underlying problem** must be identified through research and analysis.

Important Steps in Defining the Right Problem

1. Identify the Target User

First, identify the person or group experiencing the problem.

Example: College students who struggle with managing assignments.

2. Conduct User Research

Collect information through:

- Interviews
- Surveys
- Observation
- Questionnaires
- User feedback

This helps understand the user's actual experience.

3. Identify the User's Needs

Find out what users need, expect, or find difficult while performing a task.

Example: Students need an easy way to organize assignment deadlines.

4. Observe User Behaviour

Observe what users actually **do**, rather than depending only on what they say.

Example: Students frequently ask teachers about assignment submission dates.

5. Identify Facts and Patterns

Analyse the collected information and find repeated behaviours, difficulties, and common problems.

6. Find Insights and Root Causes

Ask “**Why?**” to understand the reason behind the problem. This helps identify the root cause instead of only treating the symptoms.

Example:

Problem: Students submit assignments late.

Root cause: They have difficulty managing multiple deadlines.

7. Separate the Problem from the Solution

The problem should be defined without immediately suggesting a solution.

✗ **Solution-focused:** “Students need a mobile reminder app.”

✓ **Problem-focused:** “Students need an easy way to manage multiple deadlines.”

8. Create a Clear Problem Statement or POV

Use the research and insights to create a focused, user-centered problem statement.

Example:

College students need an easy way to manage assignment deadlines because handling multiple subjects makes it difficult to remember important dates.

9. Validate the Problem

Discuss the problem with users and check whether it accurately represents their actual needs and difficulties.

10. Convert the Problem into an HMW Question

Finally, convert the defined problem into a **How Might We (HMW)** question to begin ideation.

How might we help college students manage multiple assignment deadlines easily?

Importance of Knowing the Right Problem

1. Helps identify the **actual user problem**.
2. Prevents designers from making **wrong assumptions**.
3. Keeps the design process **user-centered**.
4. Helps discover the **root cause** of a problem.
5. Provides a clear direction for **ideation**.
6. Helps avoid developing **unnecessary solutions**.
7. Increases the chance of creating a **useful and meaningful solution**.

Conclusion

Knowing the problem is an important part of the Define stage of Design Thinking. It involves understanding the **user, needs, observations, facts, patterns, and insights** before defining the problem. A correctly defined problem provides a strong foundation for **POV, HMW questions, ideation, and solution development**.